Lecture No.(12)

Cost of Production

Cost of Production

Introduction:

- This Section concern with examine how costs of production vary with output over any given period.
- The examination of the costs behavior is more important because the economic efficiency has been defined in term of cost.
- Cost to the producing firms, is the value of the factors of production used in producing it output.

Cost of Production

- In other words, cost measure the value of inputs used over any given period to produce an item.
- The expenses incurred on all inputs of production–both factor inputs and non-factor inputs are known as the cost of production. Land, labor, capital and organization are the factors of production called factor inputs. Raw materials, fuel, equipment, tools etc. are non-factor inputs.

Cost of Production

Purpose in assigning cost:

Economist interest in studding cost for many reasons:

1-to describe actual behavior of a firm.

2-to predict how the firm behavior will respond to specified change in the conditions it faces.

3-To help the firm make the best decisions in achieving its goals.

4-To evaluate how well firm use scarce resources.

Cost of Production

Short Run Costs and Long Run Costs:

- Short run is a period of time within which the firm can change its output by changing only the amount of variable factors, such as labor and raw materials etc.
- In short period, fixed factors such as land, machinery etc, cannot be changed. Costs of production incurred in the short run i.e., on variable factors are called short run costs.

Cost of Production

- The long run costs are the costs over a period in which all factors are changeable. Thus, costs of production on all factors (in the long run all factors become variable) are long run costs.

Cost of Production

Fixed and Variable Costs:

1. Fixed cost: denoted by (FC) are those that do not vary as a firm varies its output. Managers often refer to their fixed costs as overhead costs. These are costs that must be incurred in the short run even if the firm doesn't produce anything.

2. Variable cost: denoted by (VC) are those that change with output. These are the costs of variable inputs.

Cost of Production

The fixed costs include the costs of:

(a) The salaries and other expenses of administrative staff;

(b) The wear and tear of machinery (standard depreciation allowances);

(c) The expenses for maintenance of buildings;

Cost of Production

(d) Rents on leased property

(e) Interest on borrowed funds

(f) Insurance installment

Cost of Production

The variable costs include the cost of:

(a) Direct labor, which varies with output.

(b) Raw materials; and

(c) Running expenses of machinery.

Cost of Production

The sum of fixed and variable costs constitutes the total cost of production. Symbolically,

$$TC = TFC + TVC$$

Cost of Production

Average Cost:

Average cost (AC): is the total cost of producing any given output divided by the number of units produced, or the cost per unit:

$$AC = TC/Q$$

Cost of Production

Average cost may be divided two parts:

1- Average variable cost (AVC): is variable cost divided by the number of units of output produced over a given period. AVC=FC/Q

2-Average fixed cost (AFC): is fixed cost divided by the number of nits of output produced over a given period

AFC =FC/Q

Accordingly, Ac= AVC+ AFC

Cost of Production

Marginal Cost:

Marginal cost is the addition to the total cost as a result of a unit (one unit) increase in the output.

It is expressed as: $MC = \Delta TC / \Delta Q$

Where, ΔTC stands for the change in total cost and ΔTQ for total output

Cost of Production

Example:

Suppose that the following data represent cost of production for a hypothetical firm.

Cost of Production

output	FC	VC	TC	MC	ATC	AFC	AVC
0	116	0	116	-	-	-	**-**
1	116	24	140	24	140	116	**24**
2	116	44	160	20	80	58	**44**
3	116	60	176	16	586	38.6	**40**
4	116	84	200	24	50	29	**21**
5	116	124	240	40	48	23.2	**24**
6	116	180	296	50	49.3	19.3	**30**
7	116	252	368	72	52.6	16.6	**36**
8	**116**	**340**	**456**	**88**	**57**	**14.3**	**42.5**